

Brief paper

Optimal control for multi-agent persistent monitoring[☆]Cheng Song^{a,1}, Lu Liu^b, Gang Feng^b, Shengyuan Xu^a^a School of Automation, Nanjing University of Science and Technology, Nanjing 210094, China^b Department of Mechanical and Biomedical Engineering, City University of Hong Kong, Kowloon, Hong Kong

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ABSTRACT

The problem of persistent monitoring using a network of mobile agents is considered in this paper, where the goal is to drive the uncertainty of all targets to zero and patrol the whole mission domain. The uncertainty at each target point is assumed to evolve nonlinearly in time. Given a closed path, it is proved that multi-agent persistent monitoring with the minimum patrol period can be achieved by optimizing the agents' moving speed and initial locations on the path. It is also shown that the proposed approach provides a less conservative condition for persistent tasks with a constraint on the patrol period with respect to the existing works. Simulation results illustrate the effectiveness of the proposed persistent monitoring algorithm.

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1. Introduction

This paper focuses on the multi-agent persistent monitoring problem, where the goal is to patrol the whole mission domain while driving the uncertainty of all targets in the mission domain to zero. This problem arises in many applications such as environment surveillance and search and rescue. For example, in the identification of survival in disaster area using mobile robots the robots should be full aware of the regions of significant importance such as schools and sweep the entire disaster area. In this paper, trajectory planning approach is employed to address the multi-agent persistent monitoring problem. Given a closed path, periodic speed controller is designed for each agent to accomplish the persistent monitoring task, where each agent's speed only relies on its current location.

Recently, the multi-agent persistent monitoring problem has received considerable attention. In Smith, Schwager, and Rus (2012), persistent monitoring of a changing environment using mobile robots is addressed. Given a closed path, robot's speed is computed to prevent an accumulation function defined on finite locations from growing unbounded. An optimal control framework for the multi-agent persistent monitoring problem in one

dimensional mission space is presented in Cassandras, Lin, and Ding (2013), where the goal is to minimize the sum of the cumulative uncertainty at all sampling points over a fixed time horizon. In Nigam, Bieniawski, Kroo, and Vian (2012), the problem of persistent surveillance using unmanned air vehicle is considered, where the goal is to continuously survey the mission domain and to minimize the time interval between visitations to the same region.

Problems that are closely related to multi-agent persistent monitoring include multi-agent patrol and dynamic coverage. In multi-agent patrol, a network of mobile agents is required to repeatedly visit a mission domain or a set of predefined locations (Cannata & Sgorbissa, 2011; Portugal & Rocha, 2013). A criterion which is often used to evaluate the performance of patrol algorithms is idleness time (Machado, Ramalho, Zucker, & Drogoul, 2002), which is the time elapsed since a node is last observed by an agent. In Chevaleyre (2004), the idleness criterion is employed to compare two classes of patrol strategies, that is, cyclic-based strategy and partition-based strategy. Given a graph to represent the environment, optimal cooperative patrolling algorithms are proposed in Pasqualetti, Franchi, and Bullo (2012) to minimize the refreshing time and latency criteria, which are the time gap between any two visits to the same location and the time needed to inform networked mobile robots about an event occurring in the environment respectively. Frequency-based patrol is addressed in Elmaliach, Agmon, and Kaminka (2009), where the point visit frequency is optimized via constructing an optimal closed path that visits all points in the mission domain.

In dynamic coverage, the goal is to cover all points in the mission domain to the desired coverage level using a network of

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E-mail addresses: songchen@mail.ustc.edu.cn (C. Song), luliu45@cityu.edu.hk (L. Liu), megfeng@cityu.edu.hk (G. Feng), syxxu@njust.edu.cn (S. Xu).

¹ Tel.: +86 25 84303010; fax: +86 25 84303010.

mobile sensors (Hussein & Stipanovic, 2007; Song, Feng, Fan, & Wang, 2011). The main difference between multi-agent persistent monitoring and dynamic coverage lies in that the dynamic coverage task is completed when all points attain satisfactory coverage level while the persistent monitoring task would last forever. In Zhai and Hong (2013), a decentralized algorithm is developed to address the dynamic coverage problem for multi-agent systems in a region with uncertain workload density. An upper bound of the difference between the actual coverage time and the optimal time is also estimated. An awareness coverage model is proposed in Wang and Hussein (2010) which describes how “aware” networked mobile agents are of events occurring in the mission domain. In Song, Liu, Feng, Wang, and Gao (2013), the persistent awareness coverage problem for networked mobile sensors with awareness loss is addressed. When there exists a constraint on the persistent coverage period, it is shown that the persistent awareness coverage task can be accomplished if a set of linear inequalities has a solution. In Song, Feng, and Huang (2013), the optimal speed for mobile sensors along a closed path is designed to guarantee full awareness coverage of all points of interest while maximizing stability margin of awareness coverage or lifetime of a sensor network.

In contrast to the existing works by the authors in Song, Feng et al. (2013) and Song, Liu et al. (2013), given a closed path we focus on designing agents’ moving speed and initial locations on the path to guarantee multi-agent persistent monitoring with the minimum patrol period. The main contribution of this work can be summarized as follows. Firstly, in this work the uncertainty is assumed to evolve nonlinearly in time, which is different from the linear-in-time uncertainty model used in Cassandras et al. (2013), Pasqualetti, Durham, and Bullo (2012) and Smith et al. (2012). In addition, the awareness coverage model proposed in Wang and Hussein (2010) can be viewed as a special case of our uncertainty model. From the uncertainty model, a necessary and sufficient condition is derived under which the uncertainty of an arbitrary target can be driven to zero. Secondly, the patrol period of multi-agent persistent monitoring is minimized via optimizing mobile agents’ speed and initial locations sequentially. The idea is to first derive the optimal moving speed of agents for multi-agent persistent monitoring to minimize the time period during which an agent completes one cycle of the closed path. Then, an optimal initial location is designed for each agent to minimize the patrol period. Finally, it is also shown that the proposed approach provides a less conservative condition for persistent tasks with a constraint on the patrol period with respect to the existing works.

The remainder of this paper is organized as follows. In Section 2, problem formulation is given. In Section 3, trajectory planning of mobile agents is presented to achieve multi-agent persistent monitoring with the minimum patrol period. A simulation example is provided in Section 4 to illustrate the main results. Finally, Section 5 concludes the paper.

2. Problem formulation

Consider a compact mission domain D in $\mathbb{R}^2$. A network of holonomic mobile agents is required to move along a given closed path to repeatedly visit every point in the mission domain. Without loss of generality, we assume that the agents move counterclockwise on the path. Denote the mobile agents by A_i , $i \in S = \{1, 2, \dots, n\}$. Each agent possesses limited sensing capability which is described by

$$M_i(q_i, q) = \begin{cases} \frac{G_i}{r_i^4} (\|q_i - q\|^2 - r_i^2)^2 & \text{if } \|q_i - q\| \leq r_i, \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where q_i is the location of agent A_i and $r_i > 0$ is the limited sensing range (Hussein & Stipanovic, 2007).

It is noted that an arbitrary point q in the mission domain is visited by the multi-agent system in time interval $[t_s, t_f]$ if and only if the following inequality holds

$$\int_{t_s}^{t_f} \sum_{i=1}^n M_i(q_i, q) d\tau > 0. \quad (2)$$

Then, a finite period $\bar{T}$ is called as a patrol period if the following inequality holds

$$\int_{t_0}^{t_0+\bar{T}} \sum_{i=1}^n M_i(q_i, q) d\tau > 0, \quad \forall t_0 \geq 0, \forall q \in D. \quad (3)$$

Due to the definition of the patrol period, it is easy to check that if $\bar{T}$ is a patrol period for a multi-agent system, any period $\tilde{T}$ satisfying $\tilde{T} \geq \bar{T}$ is also a patrol period.

In the mission domain, there exists a finite set of stationary targets $\mathcal{P} = \{p_1, \dots, p_m\}$ to be monitored. An uncertainty function $X(p_j, t) \in [0, \infty)$ is defined on the target points, which characterizes the multi-agent system’s uncertainty about the state of each target. In this work, it is assumed that $X(p_j, t)$ increases as long as p_j is not covered by any mobile agent and evolves according to the following differential equation

$$\dot{X}(p_j, t) = \left(C(p_j) - \sum_{i=1}^n M_i(q_i, p_j) \right) g(X(p_j, t)), \quad (4)$$

$$X(p_j, 0) = X_0(p_j),$$

where $C(p_j) \geq 0$, $j = 1, \dots, m$ is a constant and $g(X(p_j, t)) = \sum_{l=1}^L a_l X(p_j, t)^l$ with $a_l \geq 0$, $l = 1, \dots, L$ being constants, $X_0(p_j)$ is the initial uncertainty of target p_j . In fact, $g(X(p_j, t))$ characterizes the impact of the current value of $X(p_j, t)$ on the change rate of $X(p_j, t)$. Note that in our model the target points’ uncertainty increases or decreases nonlinearly in time, which is different from the linear-in-time uncertainty model in Cassandras et al. (2013), Pasqualetti et al. (2012) and Smith et al. (2012). The nonlinear uncertainty model arises in many practical applications, where the changing rate of a quantity of interest is closely related to the current value of the quantity of interest. For example, during cyanobacteria break cyanobacteria concentration increases more rapidly in the regions of high cyanobacteria concentration. It is also noted that the mobile sensor network’s awareness coverage model proposed in Wang and Hussein (2010) can be regarded as a special case of the nonlinear uncertainty model.

In practice, it is often desirable to be full aware of all target points and patrol the whole mission domain as frequently as possible. In consequence, the goal of our work is to plan the mobile agents’ trajectories such that the uncertainties of all target points are driven to zero as time goes to infinity and the whole mission domain is patrolled with the minimum patrol period.

3. Optimal control for multi-agent persistent monitoring

Decoupling path planning from speed control has been established as an efficient approach to the complex trajectory planning problem (Kant & Zucker, 1986). In Elmaliach et al. (2009) and Song, Liu et al. (2013), several closed paths are constructed for agents such that when the agents move along the path perpetually all points in the mission domain are revisited persistently. In this work, instead of designing the optimal path for mobile agents we focus on addressing multi-agent persistent monitoring with the minimum patrol period via optimizing agents’ moving speed and initial locations on a given closed path.

Assumption 1. There exists a parameterized path $\Omega : [\alpha, \beta] \rightarrow \mathbb{R}^2$ with $\Omega(\alpha) = \Omega(\beta)$ such that all points in the mission domain D can be visited by an agent with limited sensing range if and only if this agent completes one cycle of the path.

This assumption often arises in the context of multi-agent persistent monitoring of large-scale mission domains, where agents' sensing range is negligible with respect to the range of the mission domain. Given the closed path Ω , we focus on controlling the speed of the mobile agents along the path to accomplish the persistent monitoring task. Let $\mathbb{R}_+$ be the set of strictly positive real numbers. For simplicity, consider periodic speed controllers $v : [\alpha, \beta] \rightarrow \mathbb{R}_+$. Then, each agent's speed only relies on its current location on the closed path, which leads to identical speed controllers for all agents. Since $v(\theta) > 0$ always holds, under the periodic speed controller each agent completes one cycle of the closed path in a period

$$T = \int_{\alpha}^{\beta} \frac{1}{v(\theta)} d\theta. \quad (5)$$

Then, one can derive a necessary and sufficient condition under which the uncertainty of an arbitrary target point in the mission domain can be driven to zero.

Lemma 1. For an arbitrary target point $p_j \in \mathcal{P}$, $\lim_{t \rightarrow \infty} X(p_j, t) = 0$ if and only if

$$\int_0^T \left(\sum_{i=1}^n M_i(q_i, p_j) - C(p_j) \right) d\tau > 0. \quad (6)$$

Proof. When no confusion arises $X(p_j, t)$ and $X_0(p_j)$ are denoted respectively by X and X_0 for brevity. From the definition of the uncertainty at a target point p_j , one has

$$\frac{dX}{g(X)} = \left(C(p_j) - \sum_{i=1}^n M_i(q_i, p_j) \right) dt.$$

Recall that $g(X(p_j, t)) = \sum_{l=1}^L a_l X(p_j, t)^l$ with $a_l \geq 0$, $l = 1, \dots, L$ being constants. By taking integration on both sides of the above equation, one has

$$\int_{X_0}^X \frac{1}{\sum_{l=1}^L a_l X(p_j, t)^l} dx = \int_0^t \left(C(p_j) - \sum_{i=1}^n M_i(q_i, p_j) \right) d\tau.$$

Let

$$F(X) = \int_{X_0}^X \frac{1}{\sum_{l=1}^L a_l X(p_j, t)^l} dx. \quad (7)$$

Firstly, consider the case where $a_1 > 0$. Then, $F(X)$ can be rewritten as

$$\begin{aligned} F(X) &= \int_{X_0}^X \frac{1}{a_1 X} dx - \int_{X_0}^X \frac{\sum_{l=2}^L a_l X^{l-2}}{a_1 \sum_{l=1}^L a_l X^{l-1}} dx \\ &= \frac{\ln X}{a_1} - \frac{\ln X_0}{a_1} - \int_{X_0}^X \frac{\sum_{l=2}^L a_l X^{l-2}}{a_1 \sum_{l=1}^L a_l X^{l-1}} dx. \end{aligned} \quad (8)$$

Note that $a_l \geq 0$, $\forall l = 1, \dots, L$. Then $a_1 \sum_{l=1}^L a_l X^{l-1} \geq a_1^2 > 0$, $\forall X \in [0, \infty)$.

$$\int_{X_0}^X \frac{\sum_{l=2}^L a_l X^{l-2}}{a_1 \sum_{l=1}^L a_l X^{l-1}} dx$$

is bounded for any $X \in [0, \infty)$. Thus, X converges to zero if and only if $F(X)$ goes to $-\infty$, that is, $\lim_{t \rightarrow \infty} \int_0^t (\sum_{i=1}^n M_i(q_i, p_j) - C(p_j)) d\tau$ goes to infinity.

Next, consider the case where $a_1 = 0$. Then, one has

$$F(X) = \int_{X_0}^X \frac{1}{X^\lambda \sum_{l=\lambda}^L a_l X^{l-\lambda}} dx, \quad (9)$$

where λ is the minimum integer satisfying $a_\lambda > 0$. Recall that every proper fraction can be decomposed into a sum of partial fraction uniquely. Then, $F(X)$ can be rewritten as

$$\begin{aligned} F(X) &= \int_{X_0}^X \frac{1}{a_\lambda X^\lambda} + \sum_{l=1}^{\lambda-1} \frac{B_l}{X^l} dx + \int_{X_0}^X \frac{\sum_{l=1}^{L-\lambda-1} C_l X^l}{\sum_{l=\lambda}^L a_l X^{l-\lambda}} dx \\ &= -\frac{1}{X^{\lambda-1}} \left[\frac{1}{a_\lambda(\lambda-1)} + \sum_{l=2}^{\lambda-1} \frac{B_l X^{\lambda-l}}{l-1} - B_1 X^{\lambda-1} \ln X \right] \Big|_{X_0}^X \\ &\quad + \int_{X_0}^X \frac{\sum_{l=1}^{L-\lambda-1} C_l X^l}{\sum_{l=\lambda}^L a_l X^{l-\lambda}} dx, \end{aligned} \quad (10)$$

where B_l , $l = 1, \dots, \lambda-1$ and C_l , $l = 1, \dots, L-\lambda-1$ can be determined by using the method of comparing coefficients for any given set of a_l , $l = 1, \dots, L$. Note that $\lim_{x \rightarrow 0^+} x \ln x = 0$. The first term in Eq. (10) converges to zero if and only if $F(X)$ goes to $-\infty$. Note also that the second term in Eq. (10) is always bounded. Thus, $X(p_j, t)$ converges to zero if and only if $\lim_{t \rightarrow \infty} \int_0^t (\sum_{i=1}^n M_i(q_i, p_j) - C(p_j)) d\tau$ goes to infinity.

It has been proved in Song, Liu et al. (2013) that $\lim_{t \rightarrow \infty} \int_0^t (\sum_{i=1}^n M_i(q_i, p_j) - C(p_j)) d\tau$ goes to infinity if and only if $\int_0^T (\sum_{i=1}^n M_i(q_i, p_j) - C(p_j)) d\tau > 0$. Therefore, the uncertainty at an arbitrary point of interest can be driven to zero if and only if $\int_0^T (\sum_{i=1}^n M_i(q_i, p_j) - C(p_j)) d\tau > 0$. The proof is thus completed. ■

Given a network of mobile agents and a closed path Ω , the patrol period of the mission domain only depends on the agents' traveling speed and their initial locations on the path. Therefore, one can minimize the multi-agent systems' patrol period of the mission domain by optimizing the agents' speed controller and initial locations. In Lemma 1, it has been shown that a multi-agent system's uncertainty of an arbitrary target point can be driven to zero if and only if condition (6) is satisfied. To transform this condition into a constraint on the agents' traveling speed, the following assumption is made.

Assumption 2. The inverse of the speed controller can be represented by

$$v^{-1}(\theta) = \sum_{k=1}^K b_k \mathcal{V}_k(\theta), \quad (11)$$

where $\mathcal{V}_k(\theta) : Q \rightarrow \mathbb{R}_+$, $k = 1, \dots, K$ are bounded continuous basis functions which are known *a priori* and $b_k > 0$, $k = 1, \dots, K$ are free parameters.

This assumption indicates that the inverse of the speed controller can be represented by a linear combination of basis functions. It has been widely employed in literature such as adaptive control (Schwager, Rus, & Slotine, 2009) due to the fact that any function defined on a compact set with certain smoothness requirements can be arbitrarily well approximated by linear combinations of basis functions (Sanner & Slotine, 1992).

Note that $\int_0^T M_i(q_i, p_j) d\tau = \int_\alpha^\beta M_i(\theta, p_j) v^{-1}(\theta) d\theta$. Then, the inequality (6) can be rewritten as

$$\sum_{i=1}^n \int_\alpha^\beta M_i(\theta, p_j) v^{-1}(\theta) d\theta > C(p_j)T.$$

Substituting Eqs. (5) and (11) into the above inequality, the necessary and sufficient condition (6) can be rewritten as

$$\sum_{i=1}^n \sum_{k=1}^K b_k \int_\alpha^\beta M_i(\theta, p_j) \mathcal{V}_k(\theta) d\theta > C(p_j) \sum_{k=1}^K b_k \int_\alpha^\beta \mathcal{V}_k(\theta) d\theta. \quad (12)$$

Then, the following lemma follows directly from Lemma 1.

Lemma 2. *Given a network of n mobile agents, the optimal speed controller $v^*(\theta)$ for the multi-agent persistent monitoring problem with the minimum period T defined in (5) can be derived by solving the following linear program*

$$\begin{aligned} & \min T \\ & \text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^K b_k \int_\alpha^\beta M_i(\theta, p_j) \mathcal{V}_k(\theta) d\theta - C(p_j) \\ & \quad \times \sum_{k=1}^K b_k \int_\alpha^\beta \mathcal{V}_k(\theta) d\theta > 0, \quad \forall j \in \{1, \dots, m\}, \end{aligned} \quad (13)$$

where $b_1, \dots, b_K$ are the optimization variables.

Denote the minimum period obtained by solving linear program (13) as T^* . Note that when a mobile agent completes one cycle of the closed path Ω all points in the mission domain can be visited by this agent. In consequence, T^* is a patrol period due to the definition (3). For simplicity, we assume that all agents are located on the path Ω initially and are ordered according to their initial locations $q_i(0)$, $i \in S$ such that when agent A_i completes one cycle of the path, it arrives at $q_{i+1}(0), \dots, q_n(0), q_1(0), \dots, q_i(0)$ sequentially. Then, one has the following lemma.

Lemma 3 (Song, Liu et al. (2013)). *Under Assumption 1, when a network of mobile agents moves along the path Ω with non-zero speed $v(\theta)$, all points in the mission domain can be covered in a period T_m which is given by*

$$T_m = \max_{i \in S} T_i, \quad (14)$$

where T_i is the time period during which an agent moves from $q_i(0)$ to $q_{i+1}(0)$ along Ω .

Given the optimal speed controller $v^*(\theta)$, we focus on optimizing agents' initial locations to minimize T_m . It is observed that $\sum_{i=1}^n T_i = T^*$ when agents move along the path with speed $v^*(\theta)$. Then, $nT_m \geq T^*$ with the equality holding if and only if T_i is identical for all $i = 1, \dots, n$. Following this idea, a procedure for computing the optimal initial locations that minimize T_m is given in

Algorithm 1. Firstly, select an arbitrary agent A_σ and let $\theta_1^* = q_\sigma(0)$, where θ_1^* denotes the optimal initial location for agent A_1 . Then, compute optimal initial location θ_i^* for agents A_i , $i = 2, \dots, n$ such that the time period during which an agent moves from θ_1^* to θ_i^* with speed $v^*(\theta)$ is $(i-1)T^*/n$. In consequence, T_i is T^*/n for all $i \in S$.

Remark 1. The idea behind the choice of the optimal initial locations for mobile agents is to equally space the agents in time along the closed path Ω . This idea is also employed in Pasqualetti et al. (2012) to derive a team trajectory with minimum refreshing time. However, in our work the patrol period is minimized via optimizing the agents' moving speed and initial locations sequentially while in Pasqualetti et al. (2012) the refreshing time is minimized via optimizing the robots' initial locations and waiting intervals for each robot at each strategically important point.

Algorithm 1 Computation of optimal initial locations for mobile agents.

- 1: Derive $v^*(\theta)$ and T^* by solving the linear program (13);
- 2: Select an arbitrary agent A_σ , let $\theta_1^* = q_\sigma(0)$;
- 3: **for all** $i \in S$ **do**
- 4: **if** $\theta_1^* + \int_0^{(i-1)T^*/n} v^*(q_\sigma(\tau)) d\tau \leq \beta$ **then**
- 5: $\theta_i^* = \theta_1^* + \int_0^{(i-1)T^*/n} v^*(q_\sigma(\tau)) d\tau$;
- 6: **else**
- 7: $\theta_i^* = \theta_1^* + \int_0^{(i-1)T^*/n} v^*(q_\sigma(\tau)) d\tau - \beta$;
- 8: **end if**
- 9: **end for**
- 10: **return** θ_i^*

Now we are at the position to present the main result.

Theorem 1. *Under Assumptions 1 and 2, a network of n mobile agents with speed controller $v^*(\theta)$ and initial locations θ_i^* , $i = 1, \dots, n$ guarantees multi-agent persistent monitoring with the minimum patrol period T^*/n .*

Proof. When each agent A_i starts from the point θ_i^* and moves along the path Ω with speed $v^*(\theta)$, one has $T_m = T^*/n$. Then, it follows from Lemma 3 that T^*/n is a patrol period. It has been shown in Lemma 2 that agents with speed controller $v^*(\theta)$ guarantee that the uncertainty of all target points is driven to zero. Therefore, mobile agents with speed controller $v^*(\theta)$ and initial locations θ_i^* guarantee multi-agent persistent monitoring with the patrol period T^*/n .

Next, we prove that given a network of n mobile agents T^*/n is the minimum patrol period by contradiction. Assume that there exists a patrol period $\bar{T}$ satisfying $\bar{T} < T^*/n$. Recall that T_i is the time period during which an agent moves from $q_i(0)$ to $q_{i+1}(0)$ along Ω . Under Assumption 1, one has $T_i \leq \bar{T}$, $i = 1, \dots, n$. Otherwise, there exist points that cannot be visited by the mobile agents every $\bar{T}$, which is inconsistent with the fact that $\bar{T}$ is a patrol period. It is noted that $T = \sum_{i=1}^n T_i$ is the time period during which an agent completes one cycle of the closed path. Apparently, T with $T \leq n\bar{T} < T^*$ is a patrol period. However, this contradicts the fact that T^* is the minimum patrol period for persistent monitoring using a mobile agent. Therefore, given a network of n mobile agents, T^*/n is the minimum patrol period. This completes the proof. ■

The following corollary is a direct result of Theorem 1.

Corollary 1. *Given a network of mobile agents and a time period $\hat{T}$, all points in the mission domain D are visited by the mobile agents*

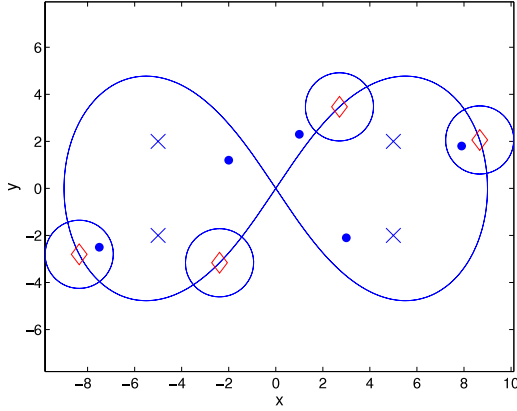


Fig. 1. Persistent monitoring task using four mobile agents.

every $\hat{T}$ and $\lim_{t \rightarrow \infty} X(p_j, t) = 0$, $\forall j \in 1, \dots, m$ if and only if T^* computed in (13) satisfies $T^* \leq n\hat{T}$.

Remark 2. It is noted that $T^*/n \leq T_m$ always holds. Therefore, Corollary 1 provides a less conservative condition for persistent tasks with a constraint on the coverage/patrol period with respect to the result in Song, Liu et al. (2013).

Remark 3. Idleness time is the time elapsing since a node is last observed by an agent and is often employed as a metric to evaluate the performance of multi-agent patrolling algorithms (Chevaleire, 2004; Machado et al., 2002; Portugal & Rocha, 2013). In this work, every point in the mission domain can be revisited by the multi-agent system every patrol period. Therefore, the patrol period can be regarded as an upper bound on the idleness over all points in the mission domain.

Remark 4. In practice, the closed path Ω often has intersections. When two agents travel in opposite directions at the intersections they would collide. To prevent agents from colliding with each other, a collision avoidance procedure based on stopping policies proposed in Soltero, Smith, and Rus (2011) can be employed. Firstly, collision zones are computed in which collisions could occur. Then, collision avoidance is guaranteed by ensuring that at most one agent occupies a collision zone at any time. It is also noted that when the collision avoidance procedure is used, the optimal initial locations of mobile agents should be recomputed with taking into account the stopping time of mobile agents.

4. Simulation results

In this section, a simulation example is provided to illustrate the performance of the proposed persistent monitoring algorithm. Fig. 1 shows four agents moving along a closed path to execute the persistent monitoring task. All agents possess uniform sensing capability, that is, $r_i = 1.6$, $G_i = 1.2$, for $i = 1, \dots, 4$. The inverse of the speed controller is parameterized as a linear combination of four Gaussian functions

$$\mathcal{V}_k = \frac{1}{2\pi} \exp \left\{ -\frac{(\tilde{q} - \mu_k)^2}{50} \right\}, \quad k = 1, \dots, 4,$$

where the locations of μ_k are marked with crosses in Fig. 1. There exist five target points to be monitored, which are denoted by solid circles as shown in Fig. 1. In Eq. (4), $g(X) = X^2 + X$ and $C(p_j)$ is set to 0.35 for all target points. The initial uncertainty of each target point for the multi-agent system is [12, 36, 60, 45, 50].

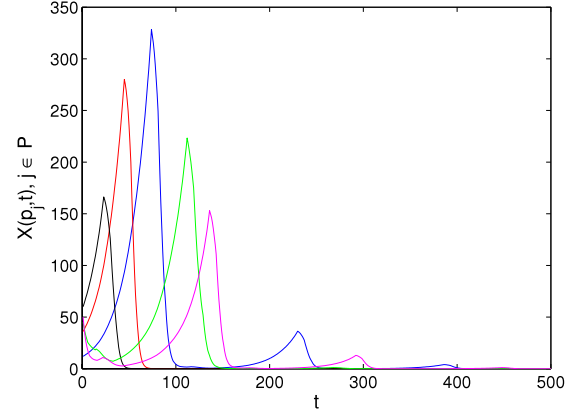


Fig. 2. Time evolution of multi-agent system's uncertainty of target points.

Table 1
Comparison of performance metric.

Speed	Initial locations	Patrol period T_m (s)
$v^*(\theta)$	θ_i^*	39.5
$\bar{v}^*(\theta)$	$\bar{\theta}_i$	127.0
$\bar{v}(\theta)$	θ_i^*	192.3
$\bar{v}(\theta)$	$\bar{\theta}_i$	629.8

Firstly, an optimal speed controller $v^*(\theta)$ for multi-agent persistent monitoring is derived by solving the linear program in (13). Fig. 2 shows the time evolution of the uncertainty $X(p_j, t)$ at each target point. It can be seen that $X(p_j, t)$ is driven to zero as time goes to infinity. Given the optimal speed controller, optimal initial locations θ_i^* , $i = 1, \dots, 4$ for each mobile agent are computed by Algorithm 1. The resulting minimum patrol period for the multi-agent system is 39.5 s.

We also compare the performance of persistent monitoring algorithms with different speed controllers and initial locations. The patrol period T_m is used as a metric to compare the performance of each algorithm. A summary of simulation results are shown in Table 1, where $\bar{v}(\theta)$ is a speed controller which satisfies the inequality (12) and $\bar{\theta}_i$, $i = 1, \dots, 4$ are agents' initial locations which are randomly deployed on the closed path. It can be seen that the persistent monitoring algorithm with $v^*(\theta)$ and θ_i^* outperforms its counterparts as stated in Theorem 1.

5. Conclusion

In this paper the problem of multi-agent persistent monitoring is formulated, where the goal is to patrol the mission domain while driving the uncertainty of all targets to zero. Persistent monitoring with the minimum patrol period is achieved via optimizing the agents' traveling speed and initial locations on a given closed path. It is noted that all targets are assumed to be stationary in this work. Future research will focus on extending the current work to the case where each target point undergoes slow drift or certain movement.

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Cheng Song received the B.Eng. degree in Automation and the Ph.D. degree in Control Science and Engineering from the University of Science and Technology of China, Hefei, China, in 2007 and 2012, respectively, and the Ph.D. degree in Mechanical and Biomedical Engineering from the City University of Hong Kong, Kowloon, Hong Kong, in 2012. He has been with Nanjing University of Science and Technology since 2012. His main research interests include mobile sensor networks and multi-agent systems.



Lu Liu received her Ph.D. degree in 2008 in Mechanical and Automation Engineering from the Chinese University of Hong Kong, Hong Kong. From 2009 to 2012, she was an Assistant Professor in the Department of Information Physics and Computing, The University of Tokyo, Japan, and then a Lecturer in the Department of Mechanical, Materials and Manufacturing Engineering, The University of Nottingham, United Kingdom. She is currently an Assistant Professor in the Department of Mechanical and Biomedical Engineering, City University of Hong Kong, Hong Kong. Her research interests are primarily in networked dynamical systems, control theory and applications and biomedical devices. She received the Best Paper Award (Guan Zhaozhi Award) in the 27th Chinese Control Conference in 2008.



Gang Feng received the B.Eng. and M.Eng. degrees in Automatic Control from Nanjing Aeronautical Institute, China in 1982 and in 1984 respectively, and the Ph.D. degree in Electrical Engineering from the University of Melbourne, Australia in 1992.

He has been with City University of Hong Kong since 2000 where he is now Chair Professor of Mechatronic Engineering. He is also Changjiang Chair Professor at Nanjing University of Science and Technology, awarded by Ministry of Education, China. He was lecturer/senior lecturer at School of Electrical Engineering, University of New South Wales, Australia, 1992–1999. He was awarded an Alexander von Humboldt Fellowship in 1997–1998, and the IEEE Transactions on Fuzzy Systems Outstanding Paper Award in 2007. His current research interests include hybrid systems and control, modeling and control of energy systems, and intelligent systems & control.

He is an IEEE Fellow, an associate editor of *IEEE Trans. on Fuzzy Systems*, and was an associate editor of *IEEE Trans. Automatic Control*, *IEEE Trans. on Systems, Man & Cybernetics, Part C, Mechatronics*, and *Journal of Control Theory and Applications*.



Shengyuan Xu received his B.Sc. degree from the Hangzhou Normal University, China in 1990, M.Sc. degree from the Qufu Normal University, China in 1996, and Ph.D. degree from the Nanjing University of Science and Technology, China 1999. From 1999 to 2000 he was a Research Associate in the Department of Mechanical Engineering at the University of Hong Kong, Hong Kong. From December 2000 to November 2001, and December 2001 to September 2002, he was a Postdoctoral Researcher in CESAME at the Université catholique de Louvain, Belgium, and the Department of Electrical and Computer Engineering at the University of Alberta, Canada, respectively. From September 2002 to September 2003, and September 2003 to September 2004, he was a William Mong Young Researcher and an Honorary Associate Professor, respectively, both in the Department of Mechanical Engineering at the University of Hong Kong, Hong Kong. Since November 2002, he has joined the School of Automation at the Nanjing University of Science and Technology as a professor.

He was a recipient of the National Excellent Doctoral Dissertation Award in the year 2002 from the Ministry of Education of China. He obtained a grant from the National Science Foundation for Distinguished Young Scholars of PR China in the year 2006. He was awarded a Cheung Kong Professorship in the year 2008 from the Ministry of Education of China.

He is a member of the Editorial Boards of the *Multidimensional Systems and Signal Processing*, the *Transactions of the Institute of Measurement and Control*, and the *International Journal of Control, Automation, and Systems*. His current research interests include robust filtering and control, singular systems, time-delay systems, neural networks, multidimensional systems and nonlinear systems.